



COVENANT UNIVERSITY

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TITLE OF EXAMINATION: B.SC EXAMINATION
COLLEGE: SCIENCE AND TECHNOLOGY
DEPARTMENT: COMPUTER AND INFORMATION SCIENCES
SESSION: 2019/2020
COURSE CODE: CSC 317
COURSE TITLE: SYSTEM ANALYSIS AND DESIGN
INSTRUCTION: ANSWER THREE (3) QUESTIONS IN ALL
TIME: 2 HOURS 30 MINUTES

SEMESTER: ALPHA
CREDIT UNIT: 2

SECTION A (Answer ALL Questions in this Section)

1. (a) A real estate company sells houses and condos (apartments). The company's listings management system stores evidence of listings. A listing contains the property address, image, year it was built, area, and price. In case of houses, people usually want to know the number of floors, size of lot, and basement characteristics. For condos, the sizes of storage and of balcony are distinguishing characteristics. The company's system associates listings with agent records. These include name, address, and employment start date. A single agent manages several listings at a time. While a listing is usually managed by a single agent, sometimes it happens that a listing is transferred from one agent to another. It is recorded when an agent starts and ends to manage a listing. Based on this description, **draw a class diagram**. You do not need to make any assumptions or to specify multiplicity. (9 marks)
- (b) The system for managing listings in a real estate company cited in Question (1a) is used by the real estate agents and managers on the daily basis. To be able to do any job with the system, any user must first have their credentials verified by the system. An agent can create a listing and then retrieve it later when needed. Some agents can update listings as well. A listing update becomes complete only after a manager approves it. Managers can get listing reports from the system. They can also remove a listing to an archive file or, when the company ceases to market a property, to a trash can (deletion). Based on this description, **draw a use case diagram**. (9 marks)
- (c) Jazzmatters is a famous Jazz festival, held in Lagos, Nigeria every year. The organizers decided to completely modernize the tickets selling system and created the following concept. From this year on, the tickets will be sold in three distinct ways: traditionally, i.e. by the two events agencies, in electronic format directly on the festival website, and through the Lagos BRT. All parties will have access to the same unique tickets database of the new system, to avoid double selling. A partnership with the BRT company needs to be set up, such that BRT can sell combi-tickets including both the festival admission fee and the bus ride to the festival venue at reduced price, from anywhere in Lagos. This way, more music fans would have easier and cheaper access to Jazzmatters. Moreover, the system will have to be extended to support not only English, but also Yoruba, Igbo, French and Hausa. Since tickets will also be sold online, Interswitch Inc. will be

contracted to provide and ensure the security of the online payment service. The Jazzmatters event manager will take care and negotiate all these details with the involved parties. Additionally, upon arrival at the festival venue, each participant has to self-check in at a touch screen terminal, which scans the barcodes on his/her ticket and issues a bracelet with an electronic chip. This can be used to load money, such that whenever (s)he wants to purchase snacks or beverages, (s)he does not have to use cash anymore, thus reducing waiting times. This measure was initiated by the program manager and will be deployed by WDL Technologies Inc. Lastly, according to the NCC regulations, the way the payment transactions are performed has to be audited by an external company at the end of the festival, since this is a public event.

Survey
Interview
Question
Observation

(i) Identify one goal, two functional requirements and two non-functional requirements in the case study. (2.5 marks)

(ii) How can you gather further requirements that help you build the Jazzmatters' new tickets selling system? Mention three (3) different requirements elicitation methods. For each, explain **why** you think the method is suitable in the context of the Jazzmatters case study, **what** stakeholders the method is appropriate for, and **how** you would implement the method in practice? (6 marks)

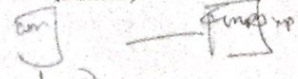
(iii) Draw a **context diagram** of the new Jazzmatters tickets selling system. Make sure you label the actors, the relationships between them as well as their relationships to the system. (6.5 marks)

(d) A manufacturing company produces products. The following product information is stored: product name, product ID and quantity on hand. These products are made up of many components. Each component can be supplied by one or more suppliers. The following component information is kept: component ID, name, description, suppliers who supply them, and products in which they are used. Create a **data model** to show how you would track this information. Show entity names, primary keys, attributes for each entity, relationships between the entities and cardinality.

Assumptions

- A supplier can exist without providing components.
- A component does not have to be associated with a supplier.
- A component does not have to be associated with a product. Not all components are used in products.
- A product cannot exist without components.

(7 marks)



SECTION B (Answer any TWO (2) Questions in this Section)

2. (a) What is the System Development Life Cycle (SDLC)? Discuss briefly the major phases in the System Development Life cycle and identify the deliverables in each phase. (10 marks)

(b) Patients call or come to the hospital to make new appointments as well as change or cancel appointments. Check to make sure the patient is a current patient and has no unpaid bills. If this is a new patient, perform "Add patient" before continuing. If the patient has unpaid bills, forward the call or patient to the account's office. If it is a change or cancellation, find the current appointment in the appointment file and cancel it. Match the patient's desired appointment times with available dates and times and schedule new appointment. Draw a Level 1 Data Flow Diagram. (5 marks)

3. (a) Discuss the feasibility aspects used in systems planning, and for each provide an example of associated risk. (5 marks)

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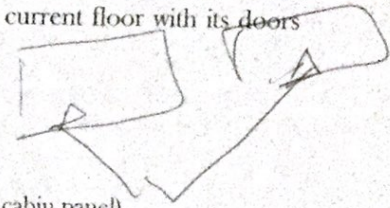
(b) When a contractor buys materials at Quality Building Supply, he takes it to the contractor checkout desk. The clerk enters the contractor name into a point of sale system (the system). The system displays the contractor record, including his current credit standing. The clerk then opens a new sales record for the contractor. Next, the clerk scans in the number of the item to be purchased into the system. The system finds the price and product name, then displays these on the screen and adds the item to the sales record. In turn, the clerk enters the quantity of the item. When all items are scanned, the clerk ends the sale. The system calculates the total and checks it against the contractor's credit limit. If the limit is not reached, the system updates the contractor's account by subtracting the amount of the sale from the credit available. If the credit limit is reached, the clerk is alerted to ask for a cash payment. If the contractor does not wish to pay in cash, the clerk cancels the sale. Successfully closed sales are finalized so that the system prints a receipt. Use all the details provided to create an **activity diagram** of the Quality Building Supply's contractor sale process.

(10 marks)

4. (a) Distinguish between a model, methodology, a tool and a technique? (5 marks)
 (b) Distinguish between rapid application development (RAD) and object-oriented design (OOD) methodologies?

(c) Each elevator has a set of **m** buttons, one for each floor. These illuminate when pressed and cause the elevator to visit the corresponding floor. The illumination is canceled when the elevator visits the corresponding floor. Each floor, except the first floor and top floor has two buttons, one to request and up-elevator and one to request a down-elevator. These buttons illuminate when pressed. The illumination is canceled when an elevator visits the floor and then moves in the desired direction. When an elevator has no requests, it remains at its current floor with its doors closed. The basic course of action for elevator scenario:

- Passenger presses floor button (that is the button on the wall).
- Elevator system detects floor button pressed.
- Elevator moves to the floor.
- Elevator doors open.
- Passenger gets in and presses elevator button (that is the button in the cabin panel).
- Elevator doors close.
- Elevator moves to required floor.
- Elevator doors open.
- Passenger gets out.
- Elevator doors close.



Draw the Sequence diagram pertaining to the elevator button.

(5 marks)

5. (a) Briefly describe the following Object-Oriented Design concepts with relevant examples:

- Class and Polymorphism
- Class and Inheritance
- Class and Composition
- Class and Aggregation
- Class and Generalization

(b) Suppose you are a System analyst working for a firm to a system development project. The system has **unclear user requirements**, a **short time schedule** and **schedule visibility**. What methodology will you recommend? Discuss this methodology, stating its advantages and disadvantages.

(5 marks)

Therived
XP Progress
(5 marks) *clear* *for* *small*

(5 marks)

10
14
24

Therived 3

(c) Suppose you are a project manager using the waterfall development methodology on a **large and complex project**. Your manager is advocating replacing the waterfall methodology with prototyping and has asked you to switch. What do you say to him? *(5 marks)*